



Group: 6

Literature Review

Machine learning techniques in Pneumonia Detection

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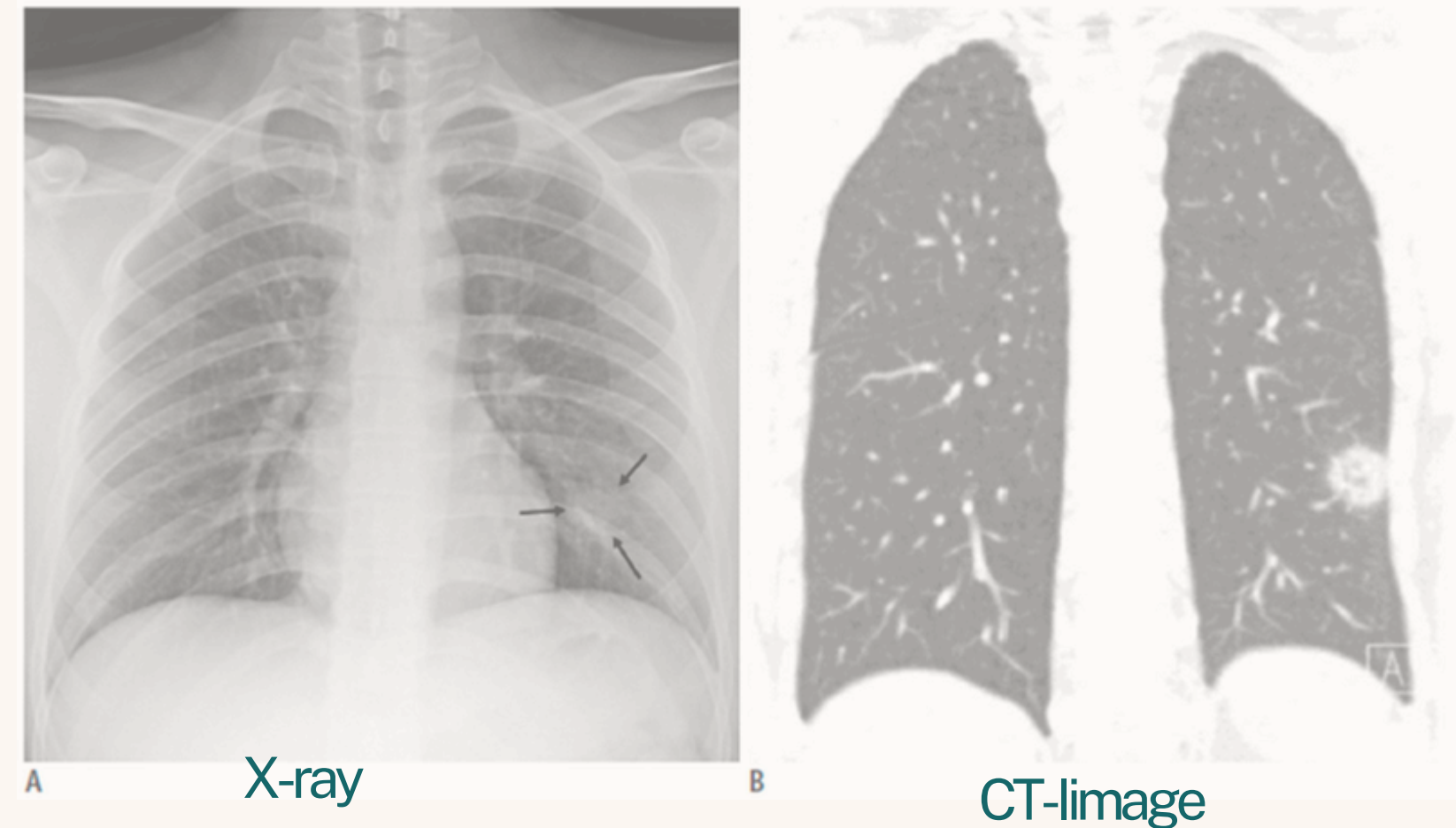


Introduction:

According to The Lancet Journal, pneumonia affects 200 million people each year, especially young children under five, and is the leading cause of death in this age group.



Scope and Objectives



With the increasing use of computer-aided technologies, chest X-ray (CXR) has become a popular technique for the early diagnosis of pneumonia. However, traditional methods struggle with complex pattern recognition, leading to potential misdiagnosis.

To overcome these challenges, we propose developing an AI-driven model that can accurately predict pneumonia without requiring human analysis of X-ray images. This approach aims to enhance diagnostic accuracy and improve patient outcomes.

Paper no 1. A federated learning framework for pneumonia image detection using distributed data (2023)

Author: Amer Kareem, Haiming Liu, and Vladan Velisavljevic (U.K)

Data: 5856 images (27% normal, and 73% as pneumonia) from UCSD.

Model: (FL) framework with Convolutional Neural Networks (CNNs): Resnet-50, AlexNet, Densenet, Inception and VGG19.

Evaluation:

	FL_Restnet_50	FL_Densenet
Train	95%	92%
Test	93%	91%
Validation	93%	91%

	FL_Restnet_50	FL_Densenet
Accuracy	95%	96%
Precision	97%	98%
Recall	96%	96%
F1-Score	0.96	0.969

Paper no 2. Automated Methods for Detection and Classification Pneumonia based on X-Ray Images Using Deep Learning

Author: Khalide El, Younes Chawki, and Ali IDRI (Morocco)

Data: X-Ray & CT dataset (5856 images) (4273 pneumomonia, and 1583 normal)

Model: CNN (Restnet50, MobileNet_V2, Inception_Resnet_V2, VGG16, VGG19, DenseNet201, Inception_V3, and Xception)

Evaluation:

	MobileNet_V2	Inception_ResNet50	ResNet50
Accuracy	96.27%	96.09%	96.61%
Precision	98.06%	98.61%	98.49%
Recall	94%	93%	94.7%
F1-Score	0.963	0.961	0.966

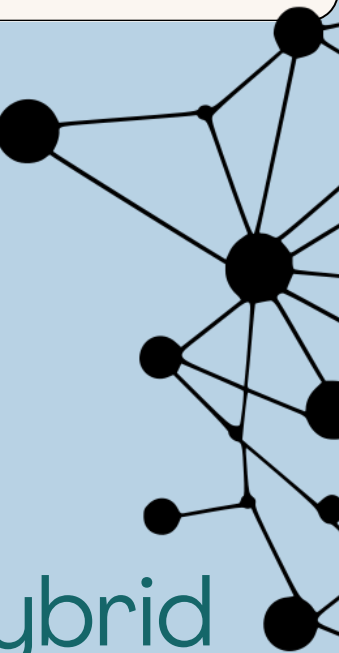
Paper 3: Identification and classification of pneumonia disease using a deep learning-based intelligent computational framework

Author: Rong Yi, Langying Tang, Yuqiu Tian, Jie Liu, Zhihui Wu

Data: Chest X-rays images.

Model: CNN-based models (Mask-RCNN, VGG16, AlexNet, and RetinaNet), Hybrid ML and DL (SVM, RNN, KNN, and MLP), and CNN with Random Forest classifier

Evaluation: The highest accuracy was 95.90%, using DL architecture with 42 convolutional layers and two dense layers. However, issues like training time and overfitting remain challenges in optimizing these models.



Paper no 4. A Deep Learning based model for the Detection of Pneumonia from Chest X-Ray Images using VGG-16 and Neural Networks

Author: Shagun Sharma, Kalpna Guleria .

Data: 6,432 images (10% covid, 24% normal, and 66% as pneumonia)
from kaggle.

Model: NN with VGG16, SVM with VGG16, KNN with VGG16, RF with VGG16 , NB with VGG16

Evaluation:

	Accuracy	Precision	Recall	F1-score
NN with VGG16	95.4%	0.954	0.954	0.954
KNN with VGG16	92%	0.922	0.919	0.969
RF with VGG16	91%	0.912	0.911	0.971
NB with VGG16	83.4%	0.872	0.834	0.834

Paper no 5. Pneumonia Image Detection: A Machine Learning Approach.

Author: Amer Kareem, Haiming Liu, Paul Sant

Data: 14,863 X-ray images.

Model: VGG16, Hybrid ML and DL (SVM, RNN, KNN, and MLP), PCA and Rule-based techniques.

Evaluation:

Algorithm	Accuracy
Rule-based techniques	96.6%
VGG-16	96.2%
PCA	95.7%

Overall and Conclusion



In conclusion, pre-trained CNN architectures, such as ResNet, DenseNet, and VGG16, demonstrated superior performance across various training methods, including federated learning (individual device training) and centralized training. These models have shown to be highly effective for pneumonia classification, indicating their potential as the best techniques for our future work in this domain.



Proposal and Future works

- Increase data for underrepresented classes to prevent dataset imbalance and reduce biased outcomes.
- Lower the prediction threshold to reduce false negatives predictions.
- Train the model using multiple image modalities, such as X-rays and CT images, to improve its generalization and robustness.
- Explore new deep learning techniques, such as YOLO and U-Net, which may enhance model performance and outcomes.



Thank
you very
much!

